

Mustain Billah Taj

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Profile

Software Engineering undergraduate with interests in backend engineering, distributed systems, AI applications, and scalable system design. Experienced in developing REST APIs, real-time applications, LLM-powered systems, and PostgreSQL-based backend services using Python and FastAPI.

Education

Bachelor of Science in Software Engineering

Islamic University of Technology (IUT)

Technical Skills

Languages: Python, Java, C++

Backend: FastAPI, REST APIs, WebSocket, WebRTC

AI/LLM: LangChain, Groq API, RAG, PGVector

Databases: PostgreSQL, MySQL, MongoDB

Tools: Docker, Git, Redis

Core CS: Data Structures, Algorithms, OOP

Projects

Stream – Self-Hosted Pay-Per-View Live Streaming Platform

Technologies: FastAPI, PostgreSQL, WebRTC, aiortc, Docker, Git, Stripe

- Real-time streaming platform supporting free and pay-per-view broadcasts with creator monetization and automated Stripe payouts.
- Custom WebRTC Selective Forwarding Unit (SFU) using `aiortc`, implementing signaling, media routing, room management, viewer authentication, and preview enforcement without third-party media servers.
- Distributed architecture separating business logic from media processing through secure ticket-based authentication, with real-time chat, moderation, and live earnings updates.

GitHub: Github Link

J_Buddy – AI Job Copilot

Technologies: FastAPI, PostgreSQL, LangChain, Groq, PGVector, Next.js, JSearch API, ARQ, Redis

- AI-powered career assistant that aggregates live job listings and delivers personalized recommendations based on user preferences.
- Retrieval-Augmented Generation (RAG) chatbot for CV analysis, interview preparation, job-specific guidance, and career-related queries.
- Intelligent career planning features including roadmap generation, learning goal recommendations, and end-to-end application tracking within a unified platform.

GitHub: Github Link

HealthHive – AI Personal Health Management Platform

Technologies: FastAPI, PostgreSQL, LangChain, Groq, PGVector, Agora, WebSocket, Next.js

- AI-driven health platform that generates personalized wellness plans, meal plans, and health milestones from user goals, medical conditions, and body metrics.
- LLM-powered coaching and semantic retrieval workflows using LangChain and PGVector, enabling context-aware health guidance while incorporating human review for safety-critical decisions.
- Integrated verified consultant onboarding, appointment scheduling, real-time video consultations with Agora, follow-up sessions, and progress tracking into a unified healthcare ecosystem.

GitHub: Github Link